

Homework — 1.2.4 Types of Programming Language

Name: _____ Date: _ **Mark:** ____ / 20

OCR H446 — set after the 1.2.4 lesson. Show your working.

Part A — This week's topic (~14 marks)

1. Explain what is meant by a **programming paradigm**, and explain why a programmer may need to choose between different paradigms. [2] (AO1)

2. State **two** features of a **procedural** programming language. [2] (AO1)

3. The following Little Man Computer (LMC) program inputs two numbers.

```

      INP
      STA first
      INP
      STA second
      SUB first
      BRP big
      LDA first
      OUT
      HLT
big   LDA second
      OUT
      HLT
first DAT
second DAT
```

(a) The user inputs **5** and then **9**. State the value output, and describe the purpose of the program for **any** two inputs. [2] (AO2)

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.....
.....
.....
(b) Describe what the instruction `BRP big` does, and explain why it is needed in this program. [2]
(AO1/AO2)

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.....
4. A processor executes the instruction `LDA 4`. The relevant contents of memory and registers are:

Address 4 holds the value 7 · Address 6 holds the value 9 · Address 7 holds the value 2 · The index register holds 2.

State the value loaded into the Accumulator when the operand 4 is interpreted using each of the following addressing modes. [4] (AO2)

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.....
.....
.....
(a) Immediate: _____

(b) Direct: _____

(c) Indirect: _____

(d) Indexed: _____

5. Explain the object-oriented concept of **inheritance** and give **one** benefit it provides to a programmer. [2] (AO1)

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Part B — Retrieval: keep it fresh (~6 marks)

6. (from 1.2.3) State **two** advantages and **one** disadvantage of the **Waterfall** methodology. [3] (AO1)

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7. (from 1.2.2) Explain the difference between **static** and **dynamic** linking of library code. [2] (AO1)

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8. (from 1.1.1) State the purpose of the **Accumulator (ACC)** register. [1] (AO1)

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